

Architecture Justification

Lab 3 — Component Modelling & Architectural Pattern Selection | Self-Service Coffee Kiosk System

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We chose a Layered Architecture for the Self-Service Coffee Kiosk System.

Architectural Choice

The system is built on a Layered Architecture with three tiers: a Presentation layer (User Interface / touch screen), a Business layer (Order Manager, Payment Service, Receipt Printer), and a Data layer (Database). Requests flow downward from the UI through the Order Manager, which coordinates payment, receipt printing, and data persistence.

Two Reasons for the Choice

- **Single, self-contained device:** The kiosk is one standalone unit with a small, fixed feature set (3 coffee types, 2 sizes, card payment, receipt). A layered style deploys cleanly on a single machine, whereas Microservices would add network calls and operational overhead that are unnecessary and overkill for one kiosk, and a distributed Client-Server split is not needed because the kiosk runs locally with its own hardware.
- **Clear separation of a sequential workflow:** The ordering process is strictly sequential (select → pay → print → store). This maps naturally onto layer-to-layer calls routed through the Order Manager, keeping responsibilities isolated, the code easy to build and maintain, and each concern (UI, business logic, data) independently testable.

Security Advantage

Payment handling is isolated inside a dedicated Payment Service in the Business layer, accessed only through the well-defined IPayment interface. The Presentation layer never touches raw card data and the Data layer stores only menu, pricing, and order records — not payment credentials. This layered isolation shrinks the attack surface and confines sensitive card processing to a single, controllable component, making secure payment handling and PCI-style controls far easier to enforce.

Performance Benefit

Because every layer runs in-process on the kiosk, communication between the UI, Order Manager, Payment Service, Receipt Printer, and Database is via fast local calls with no inter-service network latency. Layer-traversal overhead is negligible for a single-user device, so a complete order comfortably finishes within the required under-60-second target, even during busy café periods.

Component Diagram

Self-Service Coffee Kiosk System — Layered Architecture

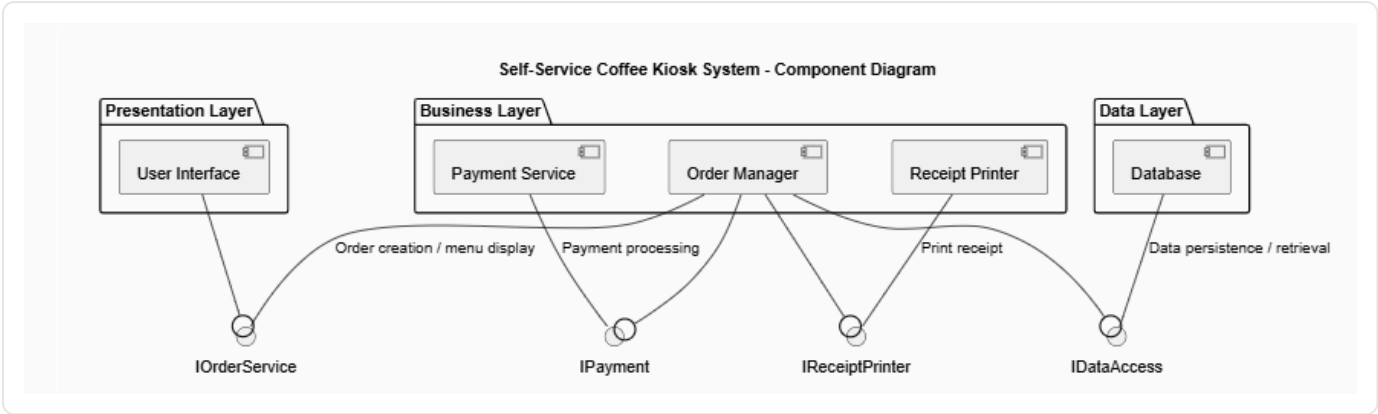


Figure 1. UML component diagram — 5 components (all as «component» rectangles) and 4 interfaces with provided (ball) / required (socket) notation.